

analysis) of that specific application, resulting in safety-relevant functions and their integrity to be performed by the PDS(SR).

4.2 Safety functions

4.2.1 Limit values

Where a safety function relies on limit value(s) for any parameter(s), the maximum tolerance(s) for the limit value(s) shall be defined.

NOTE Specification of any limit value should take into account possible exceeding of the limit value in case of violation of the limit. For example, specification of the position limit value(s) in 4.2.3.8 should take into account the maximum allowable overtravel distance(s).

A particular safety function may have one or more specified limit values, which can be selected during operation.

4.2.2 Stopping functions

4.2.2.1 General

A variety of stopping methods is available for every type of PDS.

The control requirements for initiating the stopping sequence and maintaining a hold mode upon reaching standstill are application-specific. Separate manual operations and connections to control circuits may be necessary to achieve the desired performance of the stop functions.

Any particular requirements for stopping performance should be specified by the installation designer. The following examples of stop functions are often used in practice.

4.2.2.2 Safe torque off (STO)

Power, that can cause rotation (or motion in the case of a linear motor), is not applied to the motor. The PDS(SR) will not provide energy to the motor which can generate torque (or force in the case of a linear motor).

NOTE 1 This safety function corresponds to an uncontrolled stop in accordance with stop category 0 of IEC 60204-1.

NOTE 2 This safety function may be used where power removal is required to prevent an unexpected start-up.

NOTE 3 In circumstances where external influences (for example, falling of suspended loads) are present, additional measures (for example, mechanical brakes) may be necessary to prevent any hazard.

NOTE 4 Electronic means and contactors are not adequate for protection against electric shock, and additional measures for isolation may be necessary.

4.2.2.3 Safe stop 1 (SS1)

The PDS(SR) either

- a) initiates and controls the motor deceleration rate within set limits to stop the motor and initiates the STO function (see 4.2.2.2) when the motor speed is below a specified limit; or
- b) initiates and monitors the motor deceleration rate within set limits to stop the motor and initiates the STO function when the motor speed is below a specified limit; or
- c) initiates the motor deceleration and initiates the STO function after an application specific time delay.

NOTE This safety function corresponds to a controlled stop in accordance with stop category 1 of IEC 60204-1.

4.2.2.4 Safe stop 2 (SS2)

The PDS(SR) either

- a) initiates and controls the motor deceleration rate within set limits to stop the motor and initiates the safe operating stop function (see 4.2.3.1) when the motor speed is below a specified limit; or
- b) initiates and monitors the motor deceleration rate within set limits to stop the motor and initiates the safe operating stop function when the motor speed is below a specified limit; or
- c) initiates the motor deceleration and initiates the safe operating stop function after an application specific time delay.

NOTE This safety function corresponds to a controlled stop in accordance with stop category 2 of IEC 60204-1.

4.2.3 Other safety functions

4.2.3.1 Safe operating stop (SOS)

The SOS function prevents the motor from deviating more than a defined amount from the stopped position. The PDS(SR) provides energy to the motor to enable it to resist external forces.

NOTE This description of an operational stop function is based on implementation by means of a PDS(SR) without external (for example mechanical) brakes.

4.2.3.2 Safely-limited acceleration (SLA)

The SLA function prevents the motor from exceeding the specified acceleration limit.

4.2.3.3 Safe acceleration range (SAR)

The SAR function keeps the motor acceleration and/or deceleration within specified limits.

4.2.3.4 Safely-limited speed (SLS)

The SLS function prevents the motor from exceeding the specified speed limit.

4.2.3.5 Safe speed range (SSR)

The SSR function keeps the motor speed within specified limits.

4.2.3.6 Safely-limited torque (SLT)

The SLT function prevents the motor from exceeding the specified torque (or force, when a linear motor is used) limit.

4.2.3.7 Safe torque range (STR)

The STR function keeps the motor torque (or force, when a linear motor is used) within the specified limits.

4.2.3.8 Safely-limited position (SLP)

The SLP function prevents the motor shaft from exceeding the specified position limit(s).